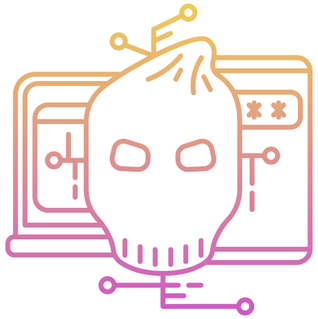
AI Based Security Detection System



TruthGuard

--Verify, Protect, Trust--

- **Course Name**: Computing Technology Innovation Project
- **Course Code**: COS30049
- **Lecturer**: Mr.Hoang (Authur) Nguyen
- **Team**: LowKenuinely (Group 4)
- **Focus:** Intelligent Spam Detection and Fake News Classification

Team Members

- Truong Ngoc Gia Hieu (Leader): Visionary & Task Delegation
- Pham Dinh Duc Anh (Vice Leader): Backend Architecture & ML Training
- Duong Nguyen Dang: UI/UX Design & Responsive Layouts
- Pahan Pathirathna: Front-end support and debugging

Problem Statement (The Digital Threat)

- **Scale:** 3.4 billion phishing emails sent daily (1.2% of global traffic).
- **Global Risk:** Misinformation ranked #1 worldwide (WEF, 2025).
- **Local Impact:** 29,000 ransomware attacks in Vietnam (2024).
- **Future Threat:** 90% of online content projected to be AI-generated by 2026.

Sources: WEF (2025), Keepnet Labs, Vietnam Cybersecurity Reports

Proposed Solution – TruthGuard

An AI-based system that processes user input and classifies it using machine learning models.

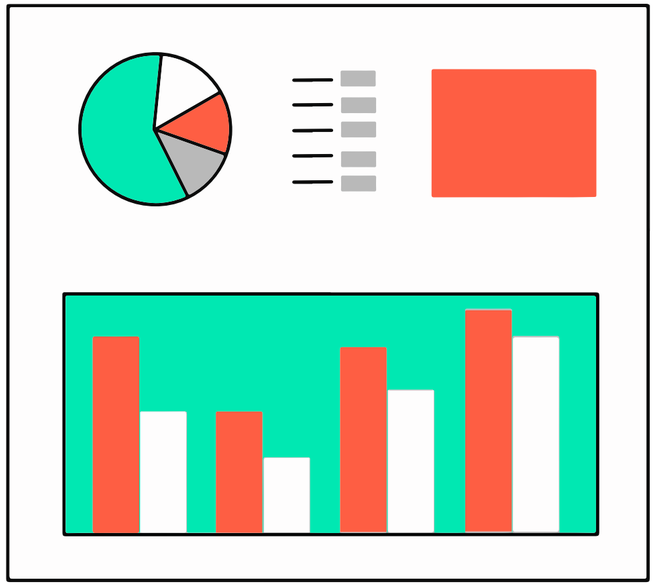
- Input Layer (**Hybrid Collection**): Text input + OCR (Optical Character Recognition) for screenshots.
- Processing Layer (**Modular Engine**): FastAPI backend routing data to specialized Neural Networks.
- Output Layer (**Interactive UI**): A React dashboard that turns complex probability into clear, color-coded results.

Objectives

- **Spam Detection:** Achieve accurate classification of spam and ham emails.
- **Fake News Identification:** Reliably detect fake or misleading news content.
- **Real-Time Performance:** Provide quick predictions with minimal delay.
- **User-Friendly Accessibility:** Maintain an interface that is simple, clear, and accessible.
- **Scalable Architecture:** Enable future expansion and integration of new features.

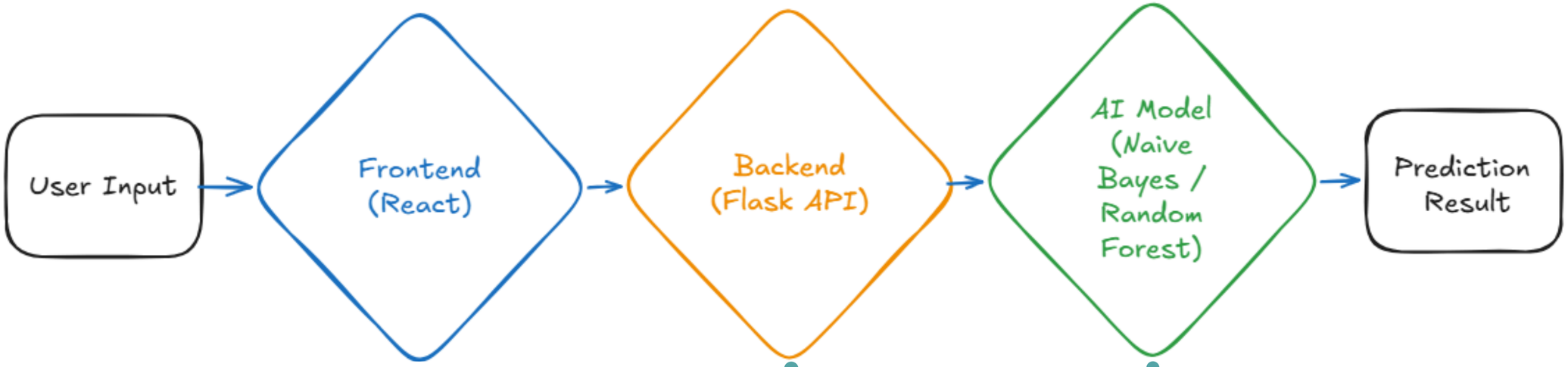
Dataset Description

- Spam/Ham email dataset
- Fake news article dataset
- Labeled data for classification
- Multiple datasets used for better accuracy



*Datasets are pre-labeled to support supervised learning.

System Architecture

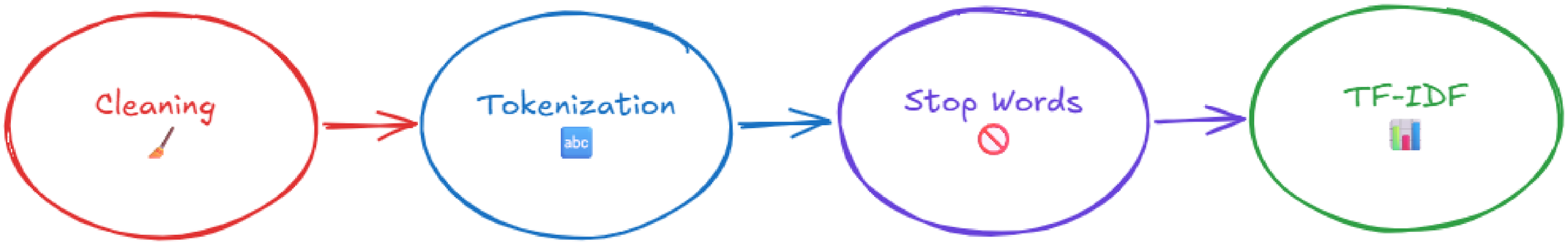


- User Interface (UI)
- Input validation
- Sends request to backend

- API processing (Flask)
- Data handling
- Communicates with AI model

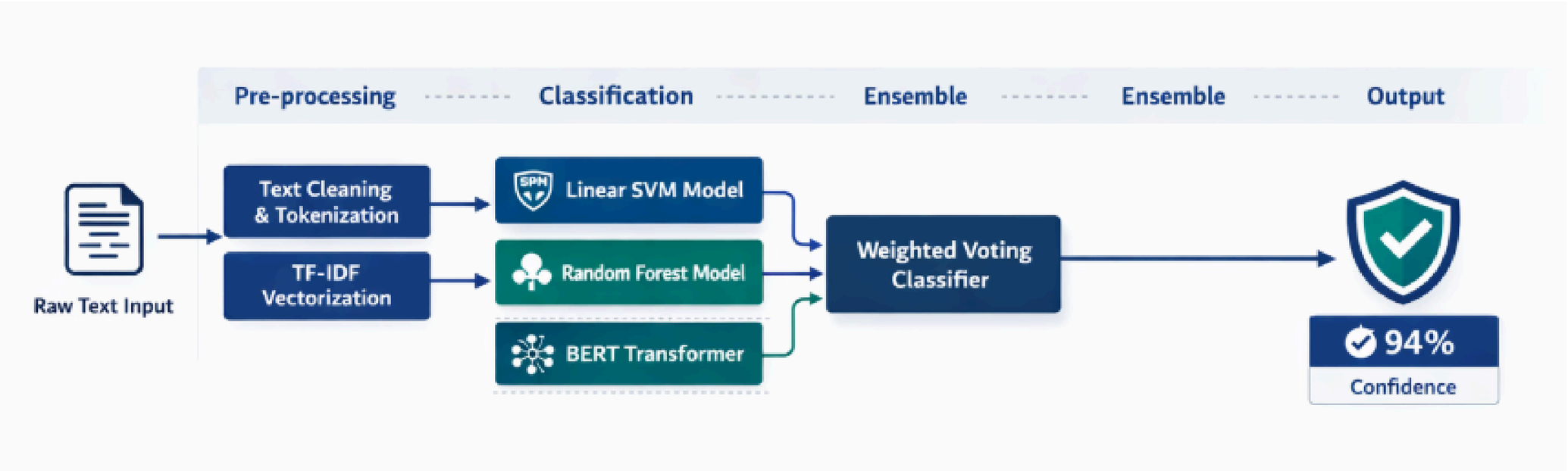
- TF-IDF Vectorization
- Classification
- Generates prediction

Data Processing

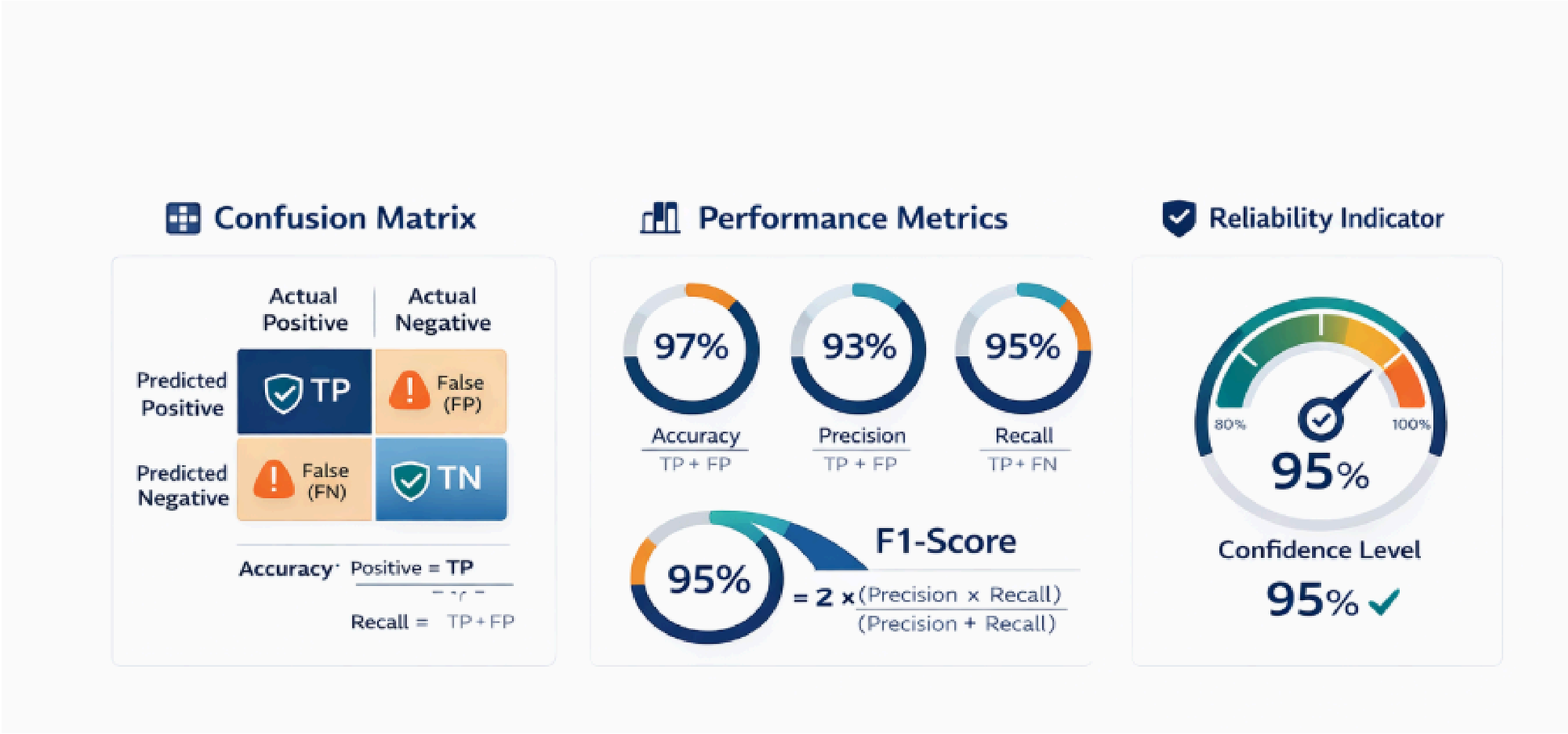


***Transforms raw text into numerical features for machine learning models.**

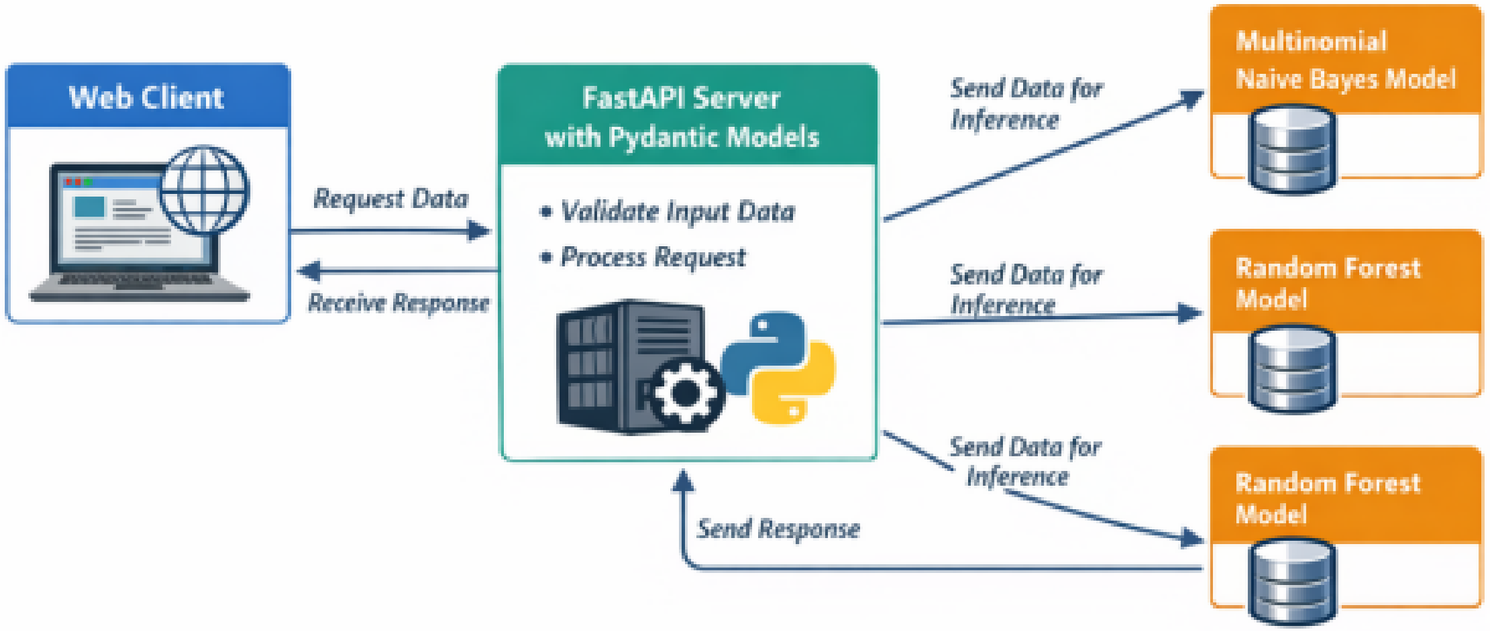
AI Algorithms



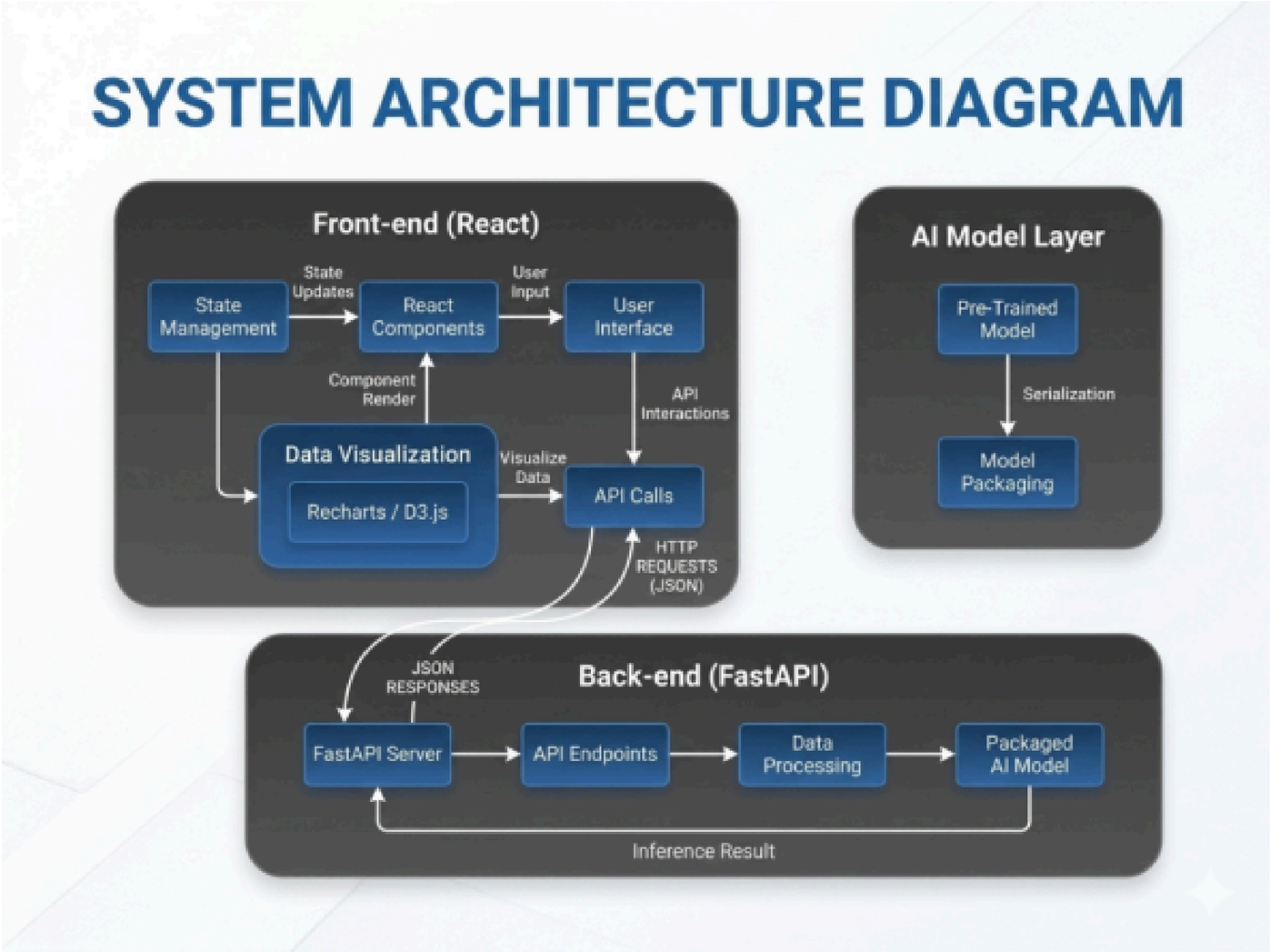
Model Evaluation



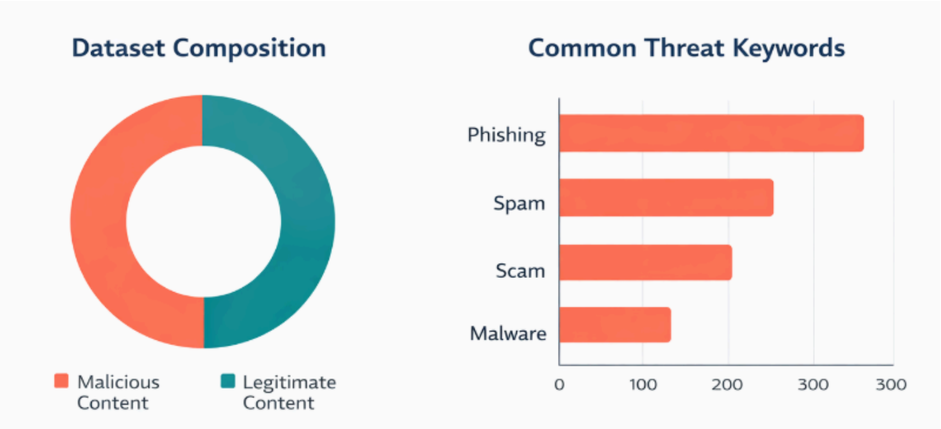
Backend Implementation

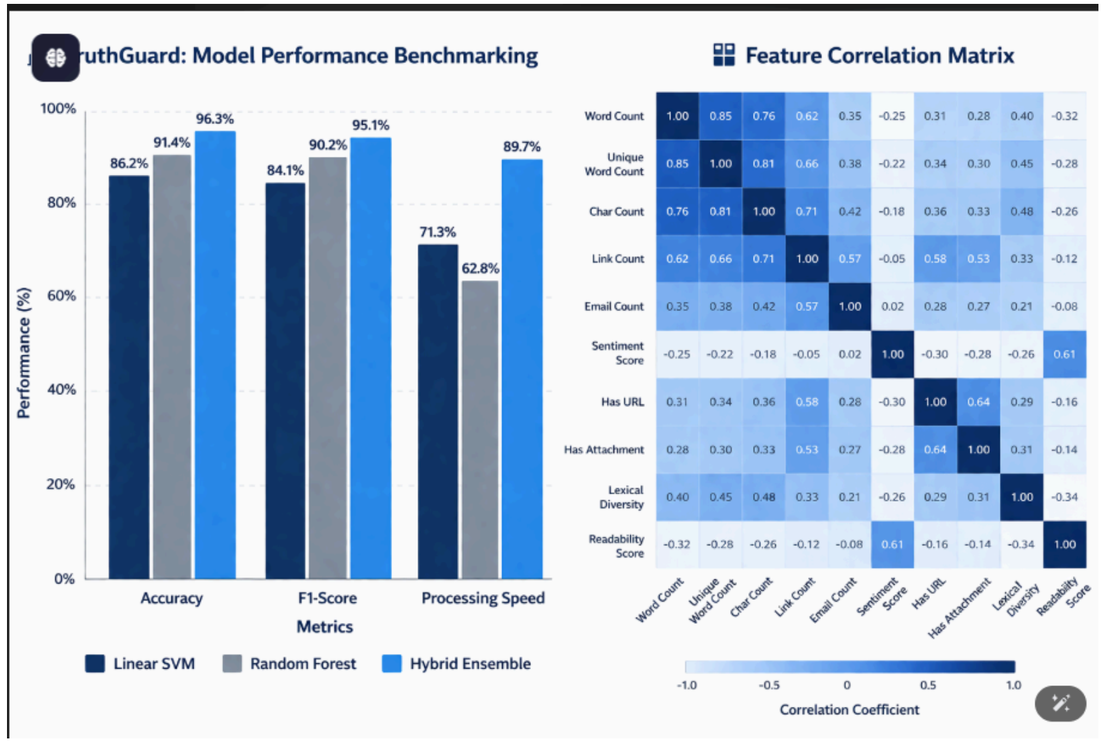


Frontend Implementation



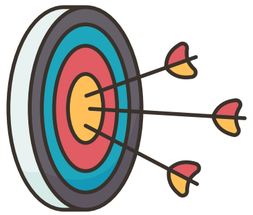
Data Visualization





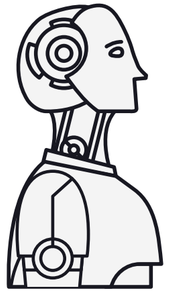
Demo

Challenges & Limitations



- **Data Quality:** AI models are highly sensitive to unstructured or noisy datasets.
- **OCR Complexity:** Stability issues when recognizing text in cluttered or low-quality images.
- **Static Nature:** Currently lacks data persistence (historical data storage for model self-optimization).
- **Language Scope:** Initially optimized for English/Vietnamese, requiring further work for global scaling.

Conclusion & Future Roadmap

- **Conclusion:** TruthGuard provides an AI-based solution for verifying digital content.

- **Future Development:**
  - Transitioning from Web to Mobile (Android/iOS).
  - Implementing Multi-language support.
  - Integrating real-time automated scaling.
  - Extensions for website